

COLLEGE OF ENGINEERING TRIVANDRUM
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
FIFTH SEMESTER B. TECH DEGREE
First Internal Exam – August 2026
PCECT502: ANALOG AND DIGITAL COMMUNICATIONS

Max. Marks: 40

Duration: 2 Hours

Part A – Answer all questions. Each Question carries 3 marks

Common scenario for Questions 1–4: Emergency rescue radio.

Following a major earthquake, conventional communication infrastructure in the affected area has been severely damaged. Rescue teams operating between collapsed buildings, temporary medical camps, and command posts therefore depend on portable radio units to exchange voice instructions and emergency information.

The communication link must remain intelligible in the presence of electrical interference, background noise, and signals from other nearby transmitters. As a rapid first solution, the engineering team considers a conventional analog radio system. They begin by evaluating AM transmission, studying how the speech information is carried by the modulated waveform and how efficiently the transmitted power and spectrum are utilised.

Because the rescue environment also contains strong amplitude disturbances from generators, machinery, and damaged electrical equipment, the team then investigates FM as a potentially more noise-resistant alternative. The receiver is designed using a superheterodyne architecture so that different incoming radio channels can be translated to a fixed intermediate frequency for reliable filtering and amplification.

After assessing the limitations of the analog link, the same rescue-worker voice channel is digitised in Part B, where PCM and related digital techniques are investigated as possible alternatives for improving robustness and transmission efficiency.

1. The team initially tests conventional AM. The message and carrier are

$$m(t) = 4 \cos(2\pi(5 \text{ kHz})t) \text{ V}, \quad c(t) = 10 \cos(2\pi(1 \text{ MHz})t) \text{ V}.$$

The modulator has amplitude sensitivity $k_a = 0.2/\text{V}$ and produces

$$s(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t).$$

Calculate the modulation index μ and the maximum and minimum envelope amplitudes. Explain how the speech information is represented by the envelope. If the message amplitude is increased to 6 V, determine the new μ and explain why an envelope detector may then produce distortion. [3] [CO1, KL3]

2. Continue with the AM signal of Question 1 for the original message amplitude. By expanding the AM waveform, identify the carrier, upper-sideband, and lower-sideband frequencies and amplitudes. Sketch the line spectrum and determine the transmission bandwidth.

If the transmitter feeds a 50Ω load, calculate the carrier power, the combined sideband power, and the total transmitted power. Determine the percentage of total transmitted power that carries the message information, and explain briefly why conventional AM is power-inefficient. [3] [CO1, KL3]

3. Because nearby electrical equipment introduces strong amplitude fluctuations, the team evaluates FM. The rescue speech has maximum frequency $f_m = 6 \text{ kHz}$. The initial peak deviation is $\Delta f_1 = 12 \text{ kHz}$ and a proposed setting uses $\Delta f_2 = 24 \text{ kHz}$.

Calculate the FM modulation index and Carson-rule bandwidth for both settings, and determine the percentage increase in bandwidth. Explain why FM is relatively insensitive to amplitude noise, how an amplitude limiter helps, and the engineering trade-off involved in increasing Δf . [3] [CO1, KL3]

4. The analog radio uses a superheterodyne receiver with fixed intermediate frequency $f_{IF} = 455 \text{ kHz}$ and must receive a station at $f_s = 1000 \text{ kHz}$.

For high-side local-oscillator injection, calculate f_{LO} and the corresponding image frequency. Show numerically that an unwanted signal at the image frequency is also converted to 455 kHz. Explain why the desired and image signals cannot be distinguished after conversion to the same IF, how the image is suppressed in practice, and why a fixed IF simplifies receiver filtering and amplification. [3] [CO1, KL3]

Part B – Answer all questions. Each question carries 14 marks.

5. PCM design for the emergency rescue radio

[14] CO1,CO2, KL4

After evaluating the analog radio in Part A, the engineering team decides to digitise the same rescue-worker voice signal. The microphone and signal-conditioning stage can produce peak excursions of up to ± 4 , V, consistent with the signal level considered in the AM system. The useful speech spectrum extends up to 6, kHz.

A PCM encoder with a uniform 8-bit quantiser covering the range -4 , V to $+4$, V is initially proposed. The available digital radio channel can support a maximum bit rate of 100, kbps. Ignore framing and synchronisation overhead unless otherwise stated.

- (i) The first prototype samples the microphone signal at $f_s = 10$, kHz.

Using the sampling theorem, determine whether this rate is sufficient for the complete speech band. Identify the range of input frequencies that can alias into the sampled baseband.

During field operation, an electrical interference component at 7.2, kHz reaches the ADC input. Determine the frequency at which this component appears after sampling. Explain why an anti-aliasing filter must therefore be placed before the ADC. **[3]**

- (ii) The sampling frequency is subsequently increased to $f_s = 16$, kHz while retaining the 8-bit uniform quantiser. Determine the number of quantisation levels and the quantisation step size. Assuming a uniform mid-rise quantiser, determine the quantisation interval containing an input sample

$$x = 1.37, \text{ V},$$

the corresponding reconstructed value, and the quantisation error.

Also determine the maximum possible magnitude of the quantisation error and explain why increasing the number of bits reduces this error. **[3]**

- (iii) For $f_s = 16$, kHz, calculate the PCM bit rate for the proposed 8-bit encoder and determine whether it can be transmitted through the available 100 kbps digital channel.

If the sampling frequency must remain at 16, kHz, determine the **maximum integer number of bits per sample** that can be supported by the channel. For this feasible resolution, calculate the corresponding number of quantisation levels and quantisation step size over the same ± 4 , V range. **[3]**

- (iv) Assume that the quantisation error of a uniform quantiser can be modelled as uniformly distributed over one quantisation interval. Calculate the quantisation-noise power for both the original 8-bit quantiser and the maximum-resolution quantiser permitted by the channel in part (iii).

For a sinusoidal input

$$x(t) = 4 \cos(2\pi f_m t) \text{ V},$$

determine its average signal power for a 1Ω reference load and hence calculate the corresponding signal-to-quantisation-noise ratio, $SQNR$, in dB for both quantisers.

Comment on whether the reduction in transmitted bit rate is obtained without a penalty in reconstructed-signal quality. **[3]**

- (v) A design engineer proposes two possible modifications:

- (a) reduce the sampling frequency while retaining more quantisation bits, or
- (b) retain the higher sampling frequency but use fewer quantisation bits.

For the 6, kHz speech bandwidth and the fixed 100, kbps channel, explain the different distortions or limitations associated with these two choices. Based on sampling, aliasing, quantisation noise, and transmission bit rate, state which quantities cannot be reduced independently when designing the PCM rescue link. **[2]**

6. Reducing the digital bit rate using DPCM and delta modulation

[14] CO2, KL3

The PCM design is reconsidered because the link should use less data while preserving intelligible speech. Since adjacent speech samples are correlated, the team evaluates DPCM and 1-bit delta modulation (DM).

- (i) **DPCM:** The predictor uses the previous reconstructed sample, $\hat{x}[n] = x_q[n - 1]$. The prediction error is quantised by a 4-bit uniform mid-rise quantiser over -0.5 V to $+0.5$ V. Determine the error-quantiser step size and the DPCM bit rate for $f_s = 10$ kHz. Calculate the percentage reduction in bit rate relative to the original 8-bit PCM system and explain why the prediction error can usually be represented using fewer bits than the original speech sample. **[3]**

- (ii) Let $x_q[0] = x[0] = 1.20 \text{ V}$, $x[1] = 1.32 \text{ V}$, and $x[2] = 1.41 \text{ V}$. Using the quantiser of part (i), calculate $e[1]$, $e_q[1]$, and $x_q[1]$; then calculate $e[2]$, $e_q[2]$, and $x_q[2]$. Explain how the receiver reconstructs each sample and why the reconstructed, rather than the original, previous sample must be used by the predictor. [4]
- (iii) **Delta modulation:** A 1-bit DM system uses $f_s = 20 \text{ kHz}$ and step size $\Delta = 50 \text{ mV}$. Determine its bit rate and the maximum input slope that can be tracked without slope-overload distortion. For an input slope of 1.2 V/ms , determine whether slope overload occurs and find the minimum step size required to avoid it at the same sampling rate. [3]
- (iv) Instead of changing Δ , the engineer doubles the DM sampling rate to 40 kHz . With $\Delta = 50 \text{ mV}$, determine the new bit rate and maximum trackable slope. Compare this bit rate with the DPCM result from part (i), and comment on whether increasing the DM sampling rate still provides a bit-rate advantage. [2]
- (v) Explain the trade-off between slope-overload distortion and granular noise as Δ is varied. For this rescue-voice link, state one practical advantage and one limitation of DM relative to DPCM. [2]

Relevant Equations

Basic relations only; use where applicable. Symbols have their usual meanings.

General signal relations

$$T = \frac{1}{f}, \quad \omega = 2\pi f,$$

$$V_{\text{rms}} = \frac{V_p}{\sqrt{2}}, \quad I_{\text{rms}} = \frac{I_p}{\sqrt{2}},$$

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt, \quad p(t) = v(t)i(t).$$

For a resistive load,

$$P = V_{\text{rms}} I_{\text{rms}} = \frac{V_{\text{rms}}^2}{R} = I_{\text{rms}}^2 R.$$

Power ratios and noise

$$\text{SNR} = \frac{P_s}{P_n},$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_s}{P_n} \right),$$

$$G_{\text{dB}} = 10 \log_{10} \left(\frac{P_o}{P_i} \right).$$

For equal impedances,

$$G_{\text{dB}} = 20 \log_{10} \left| \frac{V_o}{V_i} \right|.$$

Amplitude Modulation (AM)

$$s(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t),$$

$$\mu = k_a A_m,$$

$$P = \frac{A_p^2}{2R}.$$

Angle modulation

$$s(t) = A_c \cos \theta_i(t),$$

$$f_i(t) = \frac{1}{2\pi} \frac{d\theta_i(t)}{dt},$$

$$\theta_i(t) = 2\pi f_c t + \phi(t).$$

FM / Receiver

$$\beta = \frac{\Delta f}{f_m},$$

$$B_T \approx 2(\Delta f + f_{m,\text{max}}),$$

$$f_{IF} = |f_{LO} - f_{RF}|.$$

Basic tuned-circuit relations:

$$Q = \frac{f_0}{B}, \quad \lambda = \frac{c}{f}.$$

Sampling and discrete time

$$f_s \geq 2f_{\text{max}},$$

$$T_s = \frac{1}{f_s},$$

$$x[n] = x(nT_s),$$

$$\Omega = 2\pi \frac{f}{f_s}.$$

Quantisation

$$L = 2^n,$$

$$\Delta = \frac{V_{\text{max}} - V_{\text{min}}}{L},$$

$$|e_q|_{\text{max}} = \frac{\Delta}{2},$$

$$e_q[n] = x[n] - x_q[n].$$

General mean-square relations:

$$P_x = E\{x^2\},$$

$$\sigma_x^2 = E\{(x - \mu_x)^2\},$$

$$\text{MSE} = \frac{1}{N} \sum_{n=0}^{N-1} e^2[n].$$

PCM and digital transmission

$$R_b = n f_s,$$

$$T_b = \frac{1}{R_b},$$

$$N_{\text{bits}} = n N_{\text{samples}},$$

$$\text{SQNR}_{\text{dB}} \approx 6.02n + 1.76.$$

DPCM

$$e[n] = x[n] - \hat{x}[n],$$

$$x_q[n] = \hat{x}[n] + e_q[n].$$

A general linear predictor may be written as

$$\hat{x}[n] = \sum_{k=1}^p a_k x_q[n-k].$$

Delta Modulation

$$x_q[n] = x_q[n-1] \pm \Delta,$$

$$R_b = f_s,$$

$$\left| \frac{dx}{dt} \right|_{\text{max}} = \Delta f_s.$$

For the staircase approximation,

$$\frac{\Delta}{T_s} = \Delta f_s.$$